Build your own classifiers using both Logistic Regression and SVM to predict Titanic survival. Compare the performance of the two algorithms using the same dataset.

Titanic-Dataset.csv

Download Titanic-Dataset.csv

The Titanic dataset contains information about passengers aboard the Titanic and is commonly used for survival prediction. The dataset includes the following columns:

Survived: Survival status (0 = No, 1 = Yes)

Pclass: Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd)

Name: Passenger name

Sex: Passenger gender

Age: Passenger age

SibSp: Number of siblings/spouses aboard

Parch: Number of parents/children aboard

Ticket: Ticket number

Fare: Passenger fare

Cabin: Cabin number

Embarked: Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)

Use your model to make predictions on the test data. Calculate evaluation metrics such as accuracy and recall, and print the results.

Create a DataFrame containing three records (three passengers). Use your model to predict survival, and print the results using text descriptions such as “survived” and “not survived.”

Submission

Submit your Jupyter Notebook source code to Canvas.

Show your work to your tutor during the lab session to get marked off. This lab is assessed in Week 2 and Week 3 only. If you cannot attend, please provide supporting documentation (e.g. a medical certificate) to request an extension.